Predicting Country-Level COVID-19 Mortality Using Demographic, Economic, Healthcare, Vaccination, and Policy Factors

Springboard Data Science Career Track

Capstone Two Final Report

Prepared by

Daryl Morris

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Executive Summary

The COVID-19 pandemic produced substantial differences in mortality across countries, despite widespread access to scientific knowledge and b roadly similar public health recommendations. Understanding which country-level characteristics were most strongly associated with these differences can help governments and public health organizations better prepare for future infectious disease outbreaks.

This project investigated whether demographic, healthcare, economic, vaccination, and government policy variables could be used to predict cumulative COVID-19 mortality across countries. Publicly available data from Our World in Data, the World Bank, and the Oxford COVID-19 Government Response Tracker were integrated into a country-level analytical dataset. After data cleaning and preprocessing, three predictive modeling approaches—Linear Regression, Random Forest Regression, and Gradient Boosting Regression—were developed and compared.

Among the models evaluated, the Random Forest model demonstrated the strongest predictive performance, achieving the lowest Root Mean Squared Error (RMSE) while also producing the highest coefficient of determination (R^2). Feature importance analysis indicated that infection burden (total COVID-19 cases per million), regional differences, median age, healthcare capacity, vaccination coverage, and government response stringency were among the most influential predictors of mortality. As an exploratory extension, a simple feed-forward neural network was implemented using PyTorch. Although it did not outperform the Random Forest model, it provided valuable insight into the application of deep learning techniques for structured tabular data.

These findings suggest that COVID-19 mortality was influenced by a combination of demographic, healthcare, epidemiological, and regional factors rather than any single characteristic. Although the final model explained a meaningful portion of the variation in mortality outcomes, additional factors not captured in the available public datasets likely contributed to differences across countries.

The results demonstrate the value of machine learning methods for understanding complex public health outcomes and provide insights that may help inform future pandemic preparedness, resource allocation, and data collection efforts.

1. Introduction

The COVID-19 pandemic was one of the most significant global public health events of the twenty-first century. Although the virus spread worldwide, countries experienced dramatically different mortality outcomes. Some nations reported relatively low COVID-19 death rates, while others experienced substantially higher mortality despite access to similar scientific knowledge and many of the same public health interventions. These differences have prompted researchers and policymakers to investigate which country-level characteristics were most strongly associated with successful pandemic outcomes.

A wide range of factors have been proposed as contributors to COVID-19 mortality, including population age, healthcare capacity, economic resources, vaccination coverage, and government policy responses. However, these factors are highly interconnected, making it difficult to determine their relative importance using simple descriptive analyses alone. Machine learning methods provide an opportunity to examine these complex relationships by evaluating multiple predictors simultaneously and identifying those most strongly associated with mortality outcomes.

The objective of this project was to develop predictive models capable of estimating cumulative COVID-19 mortality across countries using publicly available demographic, healthcare, economic, vaccination, and policy data. Three regression approaches were evaluated—Linear Regression, Random Forest Regression, and Gradient Boosting Regression—to determine which modeling technique provided the strongest predictive performance. In addition to comparing model performance, the project sought to identify the country-level characteristics that contributed most strongly to differences in mortality.

The intended audience for this analysis includes public health agencies, government policymakers, healthcare planners, and global health organizations interested in understanding factors associated with pandemic outcomes. Although the analysis is observational and does not establish causal relationships, the results provide insights that may help inform future pandemic preparedness, resource allocation, and data collection efforts.

2. Project Objectives

The primary objective of this project was to investigate whether country-level demographic, healthcare, economic, vaccination, and government policy variables could be used to predict cumulative COVID-19 mortality across countries using publicly available data.

Specific objectives included:

- Integrate publicly available COVID-19, demographic, healthcare, economic, and government policy datasets into a single analytical dataset suitable for modeling.
- Explore relationships between country-level characteristics and cumulative COVID-19 mortality through exploratory data analysis and visualization.
- Develop and compare multiple predictive models, including Linear Regression, Random Forest Regression, and Gradient Boosting Regression.
- Evaluate model performance using appropriate regression metrics to identify the model that provided the strongest predictive performance.
- Identify the country-level variables that contributed most strongly to mortality predictions and interpret their relative importance.
- Provide findings and recommendations that may help inform future pandemic preparedness, public health planning, and data collection efforts.

3. Data Sources and Preparation

This project integrated multiple publicly available datasets to create a country-level analytical dataset for modeling COVID-19 mortality. Combining these sources provided information on pandemic outcomes, demographic characteristics, healthcare capacity, economic indicators, vaccination coverage, and government policy responses.

The primary source of COVID-19 data was the Our World in Data (OWID) COVID-19 dataset, which included cumulative cases, deaths, vaccination rates, booster uptake, population characteristics, and other pandemic-related variables. Because the OWID dataset contains daily observations, the data were aggregated to the country level by selecting the most recent available observations for each country. Aggregate geographic entities (such as continents and income groups) were excluded so that the analysis focused exclusively on individual countries.

Additional demographic and economic variables were obtained from the World Bank Open Data repository, including GDP per capita, life expectancy, hospital beds per thousand population, and population density. Government response measures were obtained from the Oxford COVID-19 Government Response Tracker, which provided the overall government stringency index used in the analysis.

The datasets were merged using standardized country identifiers to create a single analytical dataset. Data quality checks included identifying duplicate records, evaluating missing values, validating country matches across datasets, and removing variables with excessive missingness or limited analytical value. The resulting dataset contained one observation per country and served as the foundation for exploratory analysis, model preprocessing, and predictive modeling.

Table 1. Data Sources

Dataset	Purpose
Our World in Data (OWID) COVID-19 Dataset	COVID-19 cases, deaths, vaccinations, demographic variables
World Bank Open Data	Economic and healthcare indicators
Oxford COVID-19 Government Response Tracker	Government response stringency index

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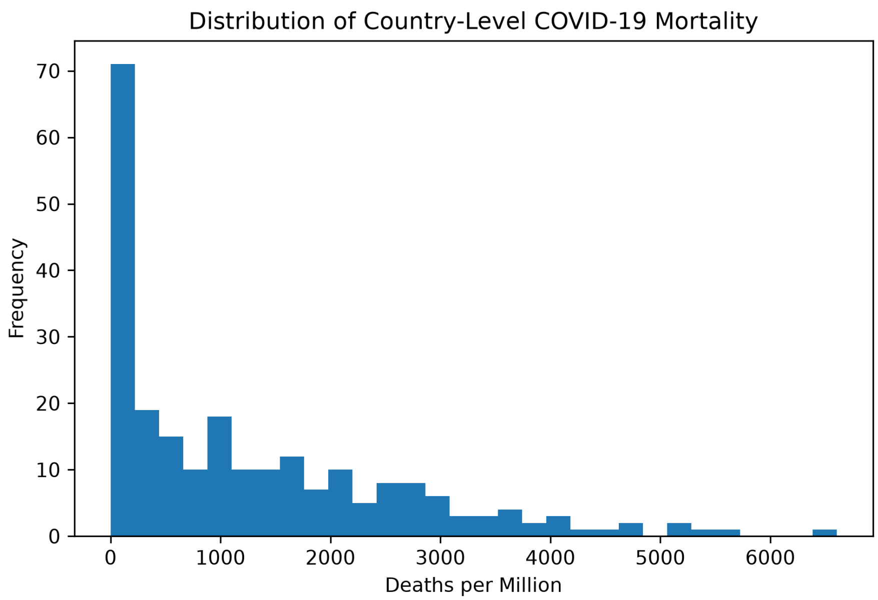
4. Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to better understand the distribution of COVID-19 mortality across countries, identify relationships between potential predictor variables and the outcome, evaluate data quality, and guide subsequent modeling decisions. Visualizations and summary statistics were used to examine the distribution of the response variable, assess correlations among predictors, identify potential multicollinearity, and evaluate the need for variable transformations.

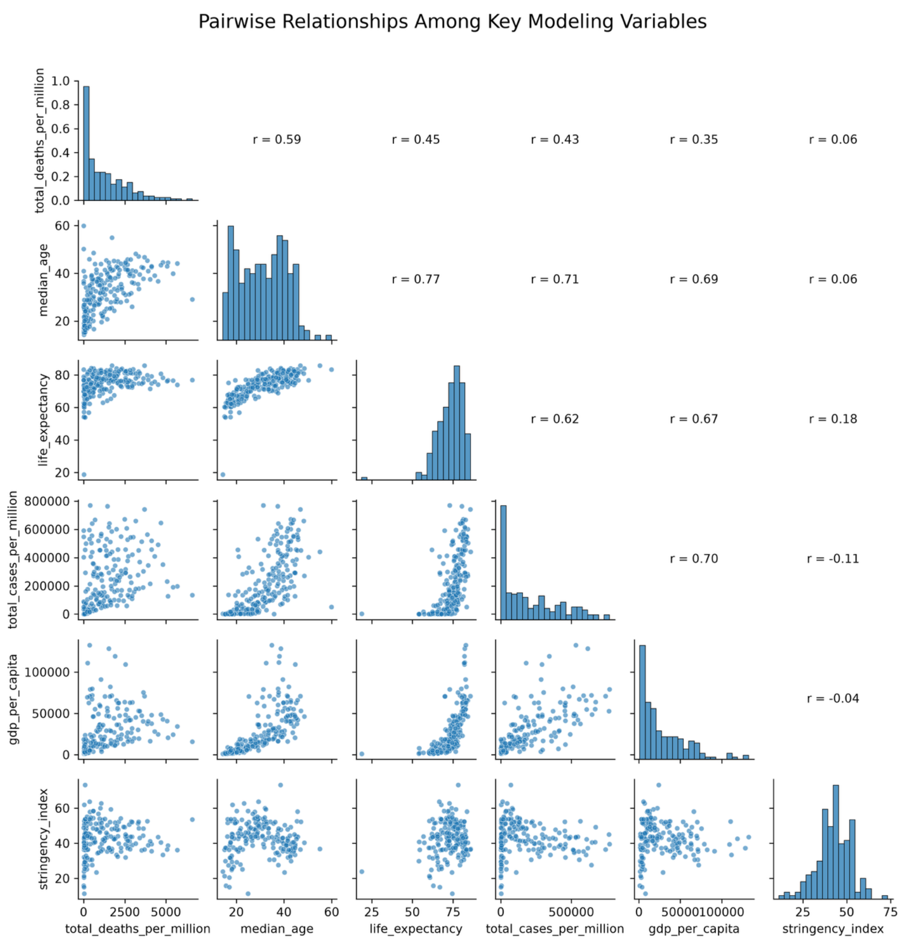
Figure 1. Distribution of Country-Level COVID-19 Mortality



The distribution of cumulative COVID-19 deaths per million population was highly right-skewed, with most countries experiencing relatively low mortality and a smaller number of countries reporting substantially higher mortality rates. This pattern suggested that predicting mortality would require models capable of handling considerable variability across countries.

To further examine relationships among the response variable and key predictors, a scatterplot matrix was created for selected variables used in the modeling process.

Figure 2. Scatterplot Matrix of COVID-19 Mortality and Selected Predictor Variables



Examination of the scatterplot matrix identified several variables that demonstrated meaningful relationships with COVID-19 mortality, including median age, life expectancy, total COVID-19 cases per million, hospital beds per thousand population, vaccination coverage, and government response stringency. Several predictors were also strongly correlated with one another, indicating that multicollinearity should be considered during model development, particularly for linear regression models.

Regional comparisons revealed substantial differences in mortality across continents. In particular, South America experienced considerably higher mortality than Asia despite similarities in several demographic and economic characteristics. These findings suggest that regional factors and infection burden may influence mortality beyond what is captured by individual demographic or economic variables alone.

Overall, the exploratory analysis identified several candidate predictors for modeling COVID-19 mortality while highlighting potential challenges related to multicollinearity, missing data, and nonlinear relationships. These findings informed the preprocessing decisions and model selection described in the following sections.

5. Data Preprocessing

Prior to model development, the analytical dataset was prepared to ensure compatibility with the machine learning algorithms evaluated in this study. Continuous predictor variables were reviewed for missing values, distributions, and scaling requirements, while categorical variables were converted into a format suitable for modeling.

The categorical variable representing continent was transformed using one-hot encoding, allowing each region to be represented as a set of binary indicator variables. To avoid perfect multicollinearity in the linear regression model, one continent was used as the reference category.

For the Linear Regression model, continuous predictor variables were standardized using the StandardScaler to place all variables on a common scale. Standardization was not required for the Random Forest and Gradient Boosting models because tree-based methods are not sensitive to differences in variable scale.

The dataset was divided into training and testing subsets using a 75%/25% split. Model development and hyperparameter tuning were performed using the training data, while the independent testing set was reserved for evaluating final model performance.

Several continuous variables, including cumulative COVID-19 deaths per million, cumulative COVID-19 cases per million, and GDP per capita, exhibited substantial right-skewness during exploratory analysis. Logarithmic transformations were evaluated to reduce skewness and improve model performance. Although the transformations produced more symmetric distributions, models using the transformed response variable did not outperform those using the original response variable. Consequently, the original outcome variable was retained for the final analyses.

6. Model Development

Three regression approaches were developed and compared to predict cumulative COVID-19 deaths per million across countries: Linear Regression, Random Forest Regression, and Gradient Boosting Regression. These models were selected to compare a traditional linear modeling approach with two ensemble tree-based methods capable of capturing nonlinear relationships and interactions among predictor variables.

Linear Regression served as a baseline model because of its simplicity and interpretability. Prior to fitting the model, continuous predictor variables were standardized to place all variables on a common scale. The model provided a useful benchmark for determining whether more sophisticated machine learning methods could reduce prediction error.

Random Forest Regression was included because it can model nonlinear relationships, automatically capture interactions among predictor variables, and is relatively robust to multicollinearity and outliers. Initial models were developed using default hyperparameters before hyperparameter tuning was performed using grid search with cross-validation. The final Random Forest model was selected based on its lowest test-set RMSE following hyperparameter tuning.

Gradient Boosting Regression was included as a second ensemble learning approach that builds trees sequentially to improve predictive accuracy. Hyperparameter tuning was also performed using grid search to identify an appropriate combination of model parameters before evaluating performance on the independent test dataset.

Because the response variable exhibited substantial right-skewness, an additional experiment evaluated models using a log-transformed response variable. Although the transformation reduced skewness, it did not improve predictive performance on the original mortality scale and was therefore not adopted for the final model.

Model performance was evaluated using root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2). Because the primary objective was to minimize prediction error, RMSE served as the principal metric for model selection, as it places greater emphasis on larger prediction errors that may be particularly important in public health applications. MAE and R^2 were reported as complementary measures of predictive performance. In addition, five-fold cross-validation was performed for the final Random Forest model to evaluate its stability and generalizability across different training subsets.

7. Results

Three regression models were evaluated using the independent test dataset to determine their ability to predict cumulative COVID-19 deaths per million across countries. Model performance was assessed using root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2). Because the primary objective was to minimize prediction error, RMSE served as the principal criterion for model selection.

Table 2. Performance Comparison of the Final Regression Models

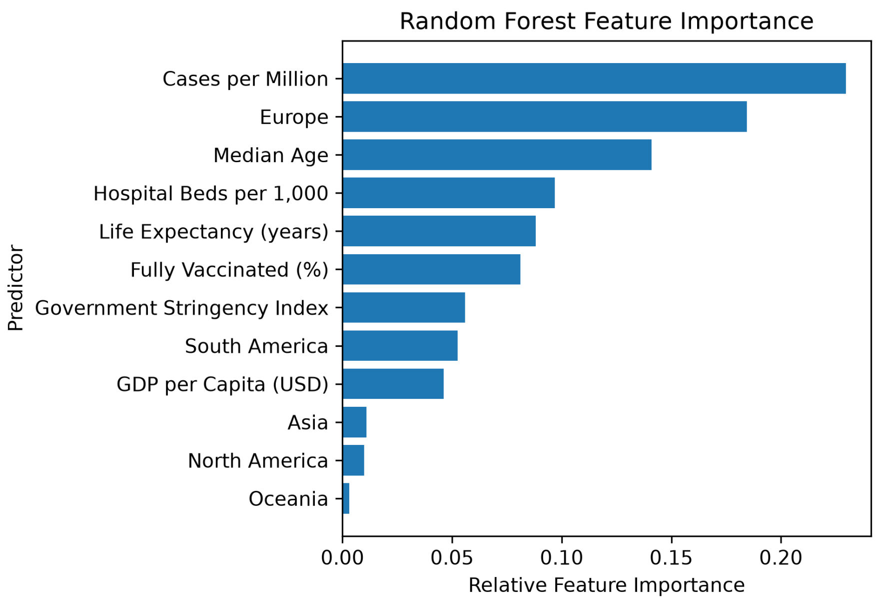
Model	RMSE	MAE	R2
Linear Regression	979.304	639.458	0.355
Random Forest	831.621	490.135	0.535
Gradient Boosting	869.06	529.634	0.492

As shown in Table 2, the Random Forest model achieved the lowest RMSE among the three models evaluated, indicating the strongest overall predictive performance on the independent test dataset. Although Gradient Boosting Regression produced competitive results, its RMSE was higher than that of the Random Forest model, indicating larger prediction errors for some countries. Linear Regression exhibited the highest prediction error, suggesting that a linear model was not sufficient to capture the complex relationships among the predictor variables. Based on these results, the tuned Random Forest model was selected as the final model for interpretation.

To evaluate the stability of the selected model, five-fold cross-validation was performed using the training dataset. The Random Forest model achieved a mean RMSE of 871.5 with a standard deviation of 174.9 across the five validation folds. These results indicate reasonably consistent predictive performance across different training subsets, suggesting that the model generalizes well despite the substantial heterogeneity present among countries.

Feature importance analysis was performed using the final tuned Random Forest model to identify the variables that contributed most strongly to the mortality predictions.

Figure 3. Feature Importance for the Final Random Forest Model



The feature importance results indicate that infection burden, demographic characteristics, healthcare capacity, vaccination measures, and regional characteristics all contributed to the model's predictions. Total COVID-19 cases per million was among the most influential predictors, while variables such as median age, life expectancy, hospital beds per thousand population, vaccination coverage, and government response stringency also made meaningful contributions. These findings suggest that no single factor explains differences in COVID-19 mortality; rather, mortality reflects the combined influence of multiple demographic, healthcare, epidemiological, and policy-related characteristics.

The predictive performance of the final Random Forest model is illustrated by comparing the observed and predicted mortality values for the independent test dataset.

Figure 4. Actual versus Predicted COVID-19 Deaths per Million for the Final Random Forest Model

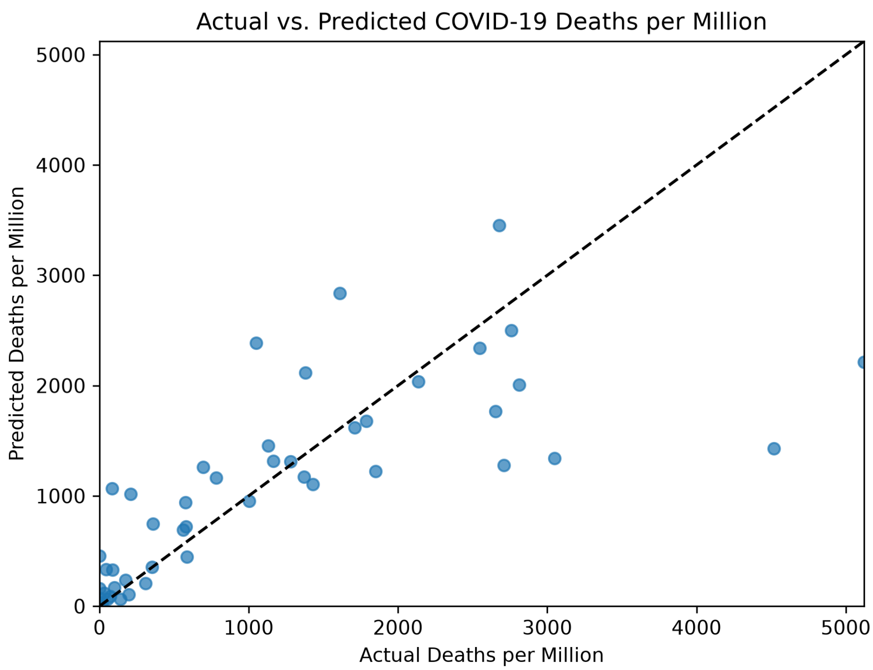


Figure 4 demonstrates that the final model captured the overall relationship between the predictor variables and observed mortality, with most predictions falling reasonably close to the line of perfect agreement. Prediction errors were larger for countries with the highest mortality rates, where the model tended to underestimate the most extreme outcomes. Overall, however, the Random Forest model provided meaningful predictive capability while acknowledging that substantial variation in country-level mortality remains unexplained by the variables included in the analysis.

8. Exploratory Deep Learning Extension

As an extension to the primary machine learning analysis, a feed-forward neural network was implemented using the PyTorch deep learning framework. The objective of this exploratory analysis was not to replace the previously developed machine learning models, but rather to compare the performance of a simple neural network with the Random Forest and Gradient Boosting approaches while gaining experience with modern deep learning techniques.

Because neural networks are generally more sensitive to the distribution and scale of the response variable than tree-based methods, the cumulative COVID-19 mortality outcome was transformed using the log1p function and subsequently standardized prior to model training. Predictor variables were likewise standardized before being converted to PyTorch tensors. The dataset was divided into the same training and testing partitions used for the earlier machine learning models to facilitate a fair comparison.

The neural network consisted of a simple fully connected feed-forward architecture with two hidden layers using ReLU activation functions. Model parameters were optimized using the Adam optimizer, with mean squared error serving as the loss function. Following training, predictions were transformed back to the original mortality scale for evaluation using the same performance metrics employed throughout the remainder of the study.

Although the neural network successfully learned meaningful relationships within the data, its predictive performance did not exceed that of the tuned Random Forest model. The Random Forest remained the strongest-performing model, producing lower prediction error and higher explanatory power on the independent test dataset. This finding is consistent with previous studies demonstrating that ensemble tree-based methods frequently outperform deep learning models on relatively small, structured tabular datasets.

Despite its lower predictive performance, the PyTorch implementation provided valuable experience with the complete deep learning workflow, including tensor preparation, neural network architecture design, iterative optimization through backpropagation, and model evaluation. The exercise also highlighted the importance of appropriate target transformations when applying neural networks to highly skewed regression problems.

Overall, this exploratory analysis demonstrated that while deep learning represents a powerful modeling framework, traditional ensemble machine learning methods remained the most appropriate approach for this particular prediction problem.

9. Conclusions

This project developed and evaluated machine learning models to predict country-level COVID-19 mortality using demographic, healthcare, economic, vaccination, and policy-related characteristics obtained from publicly available datasets. Following data integration, exploratory analysis, preprocessing, and model development, Linear Regression, Random Forest Regression, and Gradient Boosting Regression models were compared using multiple performance metrics, with RMSE serving as the primary criterion for model selection.

Among the models evaluated, the tuned Random Forest model achieved the lowest RMSE and demonstrated stable performance during five-fold cross-validation, indicating that it provided the strongest overall predictive performance. Feature importance analysis further demonstrated that COVID-19 infection burden, demographic characteristics, healthcare capacity, vaccination measures, and regional factors all contributed to mortality predictions. These findings suggest that differences in pandemic outcomes cannot be explained by any single characteristic but instead reflect the combined influence of multiple interacting factors.

Several limitations should be considered when interpreting these results. The analysis was conducted at the country level and therefore cannot account for substantial variation within individual countries. In addition, differences in reporting practices, testing availability, and mortality classification may have affected data quality across countries. Finally, because the data are observational, the results identify statistical associations rather than causal relationships.

The findings from this project may help public health organizations and policymakers better understand the types of country-level characteristics associated with pandemic outcomes. Although predictive models cannot replace epidemiological investigation or policy evaluation, they can provide useful tools for identifying higher-risk settings and supporting preparedness planning for future public health emergencies.

As an exploratory extension to the primary analysis, a simple feed-forward neural network was also implemented using the PyTorch deep learning framework. Although the neural network did not outperform the Random Forest or Gradient Boosting models, it provided valuable experience implementing a complete deep learning workflow, including data preprocessing, tensor conversion, neural network training, and model evaluation. The results suggest that, for relatively small structured tabular datasets such as those analyzed in this study, ensemble tree-based methods remain highly competitive and may offer superior predictive performance compared with simple neural network architectures.

Based on the results of this analysis, the following recommendations are offered:

1. Continue investing in standardized international public health data collection to improve the quality and comparability of future predictive models.
2. Consider multiple demographic, healthcare, epidemiological, and policy-related factors when assessing pandemic risk, rather than relying on any single indicator.
3. Explore additional data sources and modeling approaches, including longitudinal analyses and time-varying predictors, to improve predictive performance and better understand changes in mortality over the course of a pandemic.